

Towards Scalable and Real-time Markerless Motion Capture

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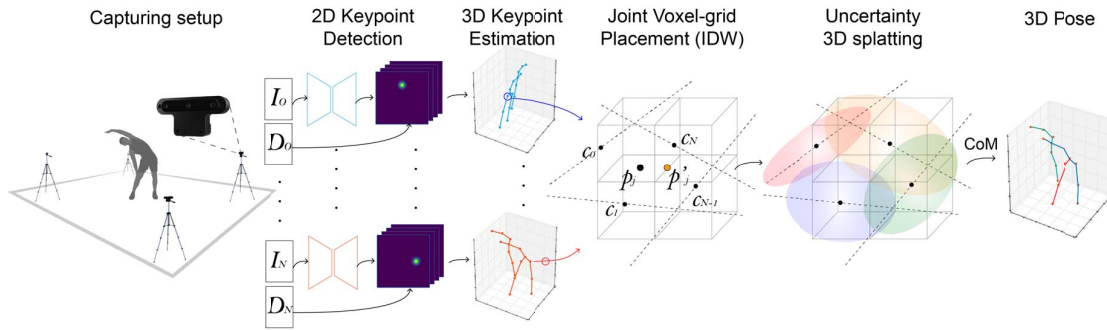


Figure 1: Our markerless motion capture method (from left to right): **i)** a multi-camera setup; **ii)** view-based 3D joint estimation based on 2D heatmap predictions; **iii)** cross-view joint's centroid estimation and voxel grid fitting; and **iv)** 3D joint estimation by computing the center of mass (CoM) of each joint's voxel grid.

ABSTRACT

Human motion capture and perception without the need for complex systems with specialized cameras or wearable equipment is the holy grail for many human-centric applications. Here, we present a scalable markerless motion capture method that estimates 3D human poses in real-time using low-cost hardware. We do so by replacing the inefficient 3D joint reconstruction techniques, such as learnable triangulation and feature splatting, with a novel uncertainty-driven approach that exploits the available depth information and the edge sensors' spatial alignment to fuse the per viewpoint estimates into final 3D joint positions.

Index Terms: Computing methodologies—Artificial intelligence—Computer vision—Motion capture; Computing methodologies—Artificial intelligence—Computer vision—3D imaging;

1 INTRODUCTION

Robust and real-time estimation of the 3D human body pose is a challenging task, but also a critical step towards digitizing, understanding, and generating human body motion. In fact, capturing body's motion is a prerequisite for many computer vision, biomechanics, and extended reality applications, such as character animation, athlete monitoring, and location-based immersive experiences.

Specifically for markerless motion capture, most recent approaches [1, 4, 6, 8] require a spatially aligned multi-camera setup and follow a two-step process; first, a 2D pose estimation model predicts the viewpoint-specific 2D joints, which are then fused with the estimates from the other viewpoints to reconstruct each joint's 3D position (e.g. via triangulation or voxel-based fusion). To overcome the

ambiguity of data-driven 2D joint estimates, the 3D reconstruction process is also learnable, a fact that hinders real-time performance when considering the entire system spanning the 2D and 3D models.

Here, we present a work-in-progress concept that aspires to overcome the need for learnable 3D reconstruction of the joint positions by instead exploiting low-cost depth sensing and modelling the uncertainty of the data-driven estimates and the systematic error of the utilized sensors in an explicit 3D reconstruction process.

2 METHOD OVERVIEW

The proposed concept, visually summarized in Fig. 1, is based on the availability of supporting depth information¹ in addition to the color input where the initial markerless 2D estimates are inferred on. Taking into account the spatially calibrated setup (Fig. 1 left-most), we are able to lift the 2D estimates to 3D positions.

Regarding 2D keypoint estimation, we can use any pretrained bottom-up model, such as the Stacked Hourglass [5] or the OpenPose [2] model, adapted (*i.e.* stack removal, quantization, etc.) to run at interactive rates on the edge sensors. This predicts heatmaps $h_i^j = f^\theta(I_i)$ for each joint j , where $f^\theta(\cdot)$ denotes the deep model with pretrained weights θ , and $I_i \in \mathbb{R}^{W \times H \times 3}$ is the color image of width W and height H captured by viewpoint $i \in \{1, \dots, N\}$.

To fuse the 2D estimations from all N viewpoints, we exploit the extrinsics of each camera and the provided depthmaps $D_i \in \mathbb{R}^{W \times H}$. However, instead of implicitly modelling the 3D reconstruction of each joint's position using learnable parameters, we employ an uncertainty-driven explicit 3D reconstruction process that is faster and scalable without requiring the same scale in the training data.

In our proposed concept, we model the uncertainty of each fused keypoint estimate with a 3D Gaussian distribution defined around its deprojected 3D position. Then, an aggregated 3D distribution is reconstructed within adaptive voxel grids defined around each joint from all estimated keypoint distributions. In more detail, the following three steps are needed: **(a)** we need to compute the centroid p^j of all viewpoint projections as an anchor point to place a $32 \times 32 \times 32$ volumetric grid V^j around the joint j , and **(b)** we compute the value

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a of each voxel v by reconstructing Gaussian distributions around the centers of the projected heatmaps. Regarding (a), we calculate p as the mean of all joint positions j across viewpoints i . The positions are lifted to 3D by extracting the corresponding coordinates from h^j (using any approach like the maximal activation, the center of mass implemented as the soft-argmax operation as in [4], or the Taylor expansion as in [9]), sampling the depth map and deprojecting the value to 3D. Then, to reduce the effect of outliers, we compute a refined centroid using Inverse Distance Weighting (IDW) [7]:

$$p' = \frac{\sum_{i=1}^N w_i c_i}{\sum_{i=1}^N w_i}, \quad (1)$$

where c_i is the center of h_i^j , and $w = \frac{1}{d(c_i, p)}$ is a weighting factor based on the distance $d(\cdot)$ between c_i and the initial centroid p (j is omitted for clarity). This way, we place the voxel grid V^j whose dimensions can be empirically defined by scaling the per viewpoint j 's bounding box, or heuristically through the j positions' standard deviation. Note that V^j is finer-grained, focusing a single joint instead of using a single grid for all joints, and is also adaptive when using each joint's position to define the physical span of V^j . Consequently, we optimize the runtime vs. accuracy trade-off, as lower voxel grids resolutions can be used for better runtime without hurting accuracy, compared to human-scale voxel grids.

After fitting V^j around p'^j , for step (b) we compute the value of each $v_k, k = \{1, \dots, K\}$ where K is the number of voxels in V^j as:

$$a_{v_k} = \sum_{i=1}^N e^{-\left(\frac{(x_k - x_i^j)^2}{2\sigma_{x_i^j}^2} + \frac{(y_k - y_i^j)^2}{2\sigma_{y_i^j}^2} + \frac{(z_k - z_i^j)^2}{2\sigma_{z_i^j}^2} \right)}. \quad (2)$$

The standard deviations crucially model the uncertainty –which comes in the form of continuous distributions– of each keypoint by jointly considering the uncertainty of the data-driven keypoint estimation ($\sigma_{x_i^j}, \sigma_{y_i^j}$), and the uncertainty of the depth measurement ($\sigma_{z_i^j}$). For the latter, we employ distance- or sensor-based modelling, while for the former we define them as a function of the keypoint estimation (e.g. the sampled confidence provided by each h^j) and the depth value z , in order to lift them to physical scale. As a last step, from the resulting V^j , we extract the center of mass that constitutes each joint's 3D position (see Fig. 1 initial estimation V), without requiring any learning process or parameters.

3 DISCUSSION

The proposed markerless MoCap method is capable of quickly predicting the 3D joints in real-time without any actual learning (see Fig. 2). It relies on pretrained 2D pose estimation models and exploits depth sensing to fuse their 3D estimates. This discards inefficient 3D learning techniques like triangulation or feature splatting but suffers from systemic (i.e. depth) and algorithmic (i.e. data-driven model) noise. To overcome these, we model the uncertainty of these processes in an explicit fusion step that smoothly extracts the keypoints in the physical coordinate system where the sensors are spatially aligned.

Nonetheless, given that we basically splat 2D keypoint heatmaps in 3D space, we can choose to further improve these estimates using a shallow and low-resolution – thus efficient- voxel-to-voxel network, such as V2V-PoseNet presented in [3]. Given a dataset that offers the final 3D joint positions as ground truth (e.g. the multiview Kinect dataset presented in [10]), our approach is end-to-end trainable (including or not the 2D pose estimation models) using for instance the $L2$ loss $\mathcal{L} = \sum_j \|y^j - y_{gr}^j\|_2^2$ to supervise the full pipeline, where y^j is the 3D joint coordinates predicted by V2V-PoseNet and y_{gr}^j the corresponding ground truth.



Figure 2: Qualitative examples of predicted (orange) and ground truth (cyan) 3D poses using our markerless MoCap method with in-house captured data.

Apart from any potential performance improvements, overcoming triangulation or 3D feature splatting with our uncertainty-driven approach enables the smooth scaling of the method for more cameras, as the only extra computational cost is the additional heatmap $h_{i_{new}}$ that participates in the efficient centroid p and voxel v_k value calculations. As a next step, we plan to further investigate the accuracy vs. performance trade-off of our method in single and multi-person scenarios, and explore the gains of end-to-end, or 3D only training.

4 CONCLUSION

In this paper we briefly presented a work-in-progress markerless MoCap concept, which leverages the recent advances in 3D pose estimation and motion capture research domains and is based on design principles that will enable real-time performance and viewpoint scalability, both from a performance point of view, as well as an accuracy/robustness one. In particular, we discussed how we leverage state-of-the-art 2D keypoint estimation models to predict joints per viewpoint, how we take advantage of a calibrated multi-camera setup, as well as how we model the introduced depth-based multi-view uncertainty to replace the computationally expensive triangulation and allow for the exploitation of lower-cost hardware.

ACKNOWLEDGMENTS

A. Chatzitofis and S. Thermos acknowledge financial support by the RIF through the KinesisVision project (PRE-SEED/0719/0119).

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